

Latent Structure in Big Five Personality Ratings: PCA Baseline and Factor Analysis with Robustness Checks

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1 Introduction

Personality is commonly summarized by the Big Five framework, which describes individual differences through five broad latent traits [1]. This report analyzes responses to the 50 item IPIP Big Five questionnaire [2], with each item scored from 1 to 5, to determine whether the correlation structure among the items can be represented by a low dimensional latent structure and, if so, how that structure should be estimated and interpreted.

This project emphasizes methodological rigor rather than simply reproducing the expected five factor structure. Although the Big Five provides a clear theoretical benchmark, empirical results may depend on choices such as the correlation measure for ordinal items, the criterion for selecting dimensionality, and the rotation strategy. The goal is therefore to recover an interpretable latent structure and assess how sensitive it is to these reasonable alternatives.

This report analyzes the Big Five IPIP dataset (50 Likert items, 1–5) to understand its underlying low dimensional structure and how that structure depends on modeling choices. Here are three research questions I will address in the later analysis:

- **Latent Dimensionality:** Does the item correlation structure support a parsimonious five-dimensional latent space as theoretically expected, and how do data-driven criteria such as parallel analysis and scree evidence compare in determining the optimal number of factors.
- **Methodological Suitability:** Given the distributional properties and measurement scale of the IPIP items, is a common-factor model or an independent component approach more appropriate for recovering interpretable latent personality traits?
- **Structural Robustness:** To what extent are the resulting latent dimensions stable across different modeling choices, specifically regarding the type of correlation matrix used (Pearson vs. polychoric) and the strategy for factor rotation?

The analysis starts with basic EDA of item distributions and the correlation structure. I then run PCA as a baseline and fit factor analysis to interpret the latent traits (with rotation). Finally, I check robustness to key modeling choices, including Pearson vs polychoric correlations and alternative factor specifications.

2 Data Overview and Preprocessing

2.1 Dataset description

The analysis uses the cleaned Big Five Personality dataset based on the 50 item IPIP Big Five Factor Markers. The dataset contains responses from 19,718 respondents and includes 50 questionnaire items, structured as 5 domains with 10 items per domain. The five domains are Extraversion (E1–E10), Neuroticism (N1–N10), Agreeableness (A1–A10), Conscientiousness (C1–C10), and Openness to Experience (O1–O10).

All items are recorded on a 1–5 Likert scale from “disagree strongly” to “agree strongly.” The data provided for this project contains no missing values in the cleaned dataset. The dataset used here also retains only the 50 personality item columns with complete responses, excluding demographic variables available in the original source

2.2 Questionnaire coding and analysis scale

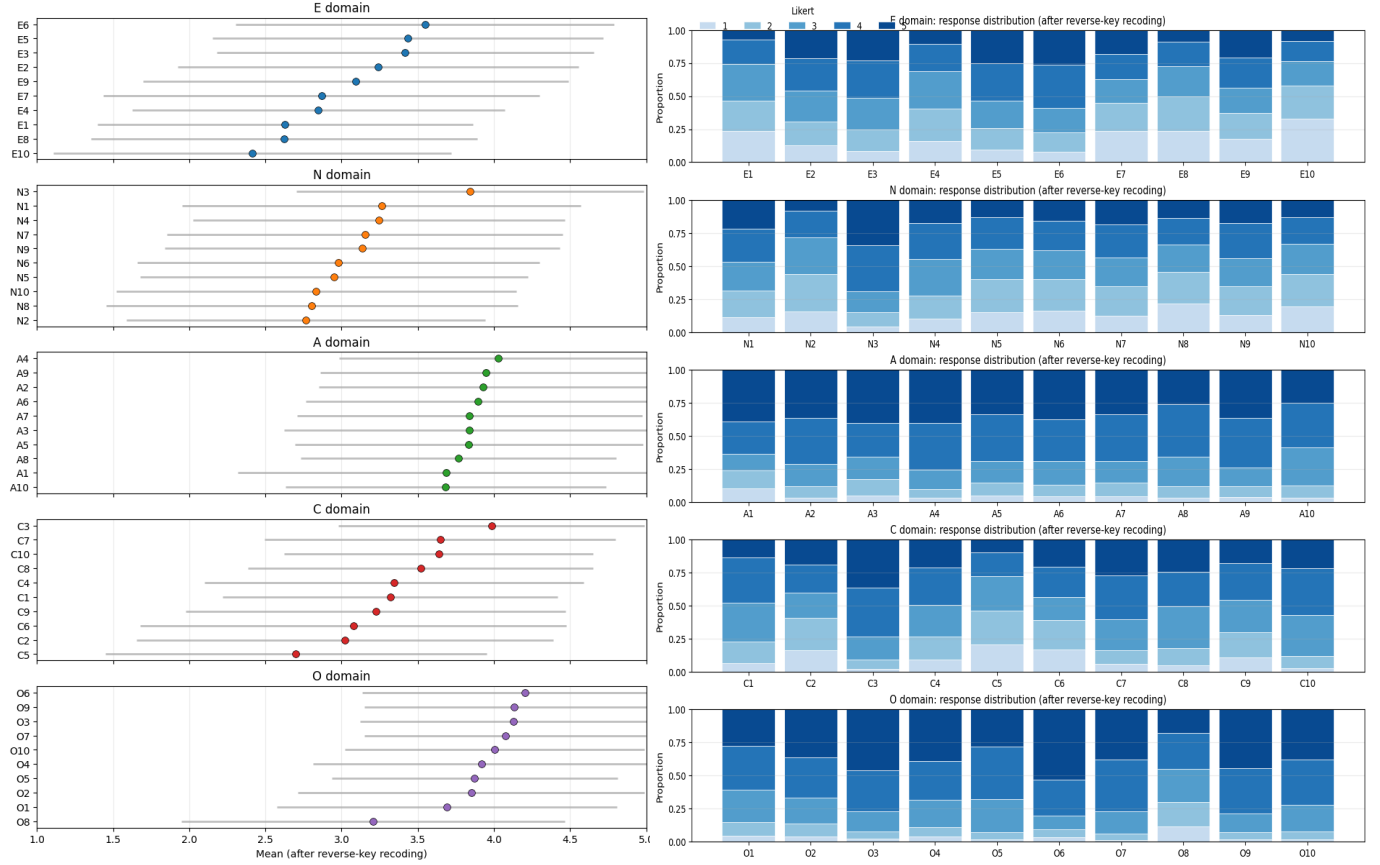
Items were presented in an interleaved order across domains (e.g., E1, N2, A1, C1, O1, E2, ...) to reduce response bias, and some items are reverse keyed. These reverse keyed items were recoded using $x \mapsto 6 - x$ so that higher scores consistently indicate more of the corresponding trait; this prevents correlations and loadings from being driven by item wording direction rather than the intended latent trait. Because all items share the same 1–5 scale, no additional rescaling was needed. Subsequent analyses are based on inter item correlation structure rather than raw marginal scales. As a descriptive baseline, PCA is performed on the Pearson correlation matrix of the standardized item responses, whereas the primary factor analysis is fit to the polychoric correlation matrix to better reflect the ordinal measurement scale.

2.3 Univariate distribution checks

Before any multivariate modeling, two univariate summaries were used to check basic scale usage and variability after reverse key recoding.

First, Figure 1(a) summarizes item means and variability after reverse key recoding. Across all 50 items, mean responses range from 2.41 (E10) to 4.21 (O6), while item standard deviations range from 0.92 (O7) to 1.43 (E4). At the domain level, Agreeableness and Openness items have the highest average endorsement, with domain level mean averages of 3.84 (A) and 3.91 (O), whereas Extraversion items are lowest on average at 3.01. Variability is broadly comparable across domains, though Openness shows the smallest average dispersion (average SD 1.05) and Extraversion the largest (average SD 1.29), suggesting some domains are more tightly clustered in responses than others.

Second, Figure 1(b) shows the full Likert response profiles and highlights domain specific skew. Averaging across items within each domain, the proportion of responses in categories 4 or 5 is 0.69 for Openness and 0.68 for Agreeableness, indicating strong concentration toward agreement. In contrast, Extraversion is much more balanced, with 0.40 in categories 4 or 5 and 0.38 in categories 1 or 2. At the item level, endorsement of the top category varies substantially: for example, the proportion of “5” responses is 0.53 for O6 but only 0.08 for E10. These patterns motivate focusing subsequent PCA and factor analysis on the inter item correlation structure rather than relying on marginal summaries alone.



(a) Item means (± 1 SD) by domain after reverse key recoding, showing between item differences in average responses and variability within each Big Five domain.

(b) Likert response distributions by domain after reverse key recoding, showing the full 1–5 response profiles for each item and highlighting differences in response skew across items.

Figure 1: Univariate summaries of the Big Five items after reverse key recoding. Panel (a) reports item level means with ± 1 SD, and panel (b) reports item level response proportions on the 1–5 Likert scale. Together these plots provide a baseline check of item scale usage and variability prior to correlation based PCA and factor analysis.

3 Assumption Checks and Factorability

3.1 MVN assessment

To assess the multivariate normality (MVN) assumption, Mardia’s tests were applied to a random subsample of $n = 3000$ observations across the $p = 50$ standardized, reverse-key-recoded items. Both the multivariate skewness and kurtosis tests strongly rejected MVN (skewness: $b_{1,p} = 104.67$, $\chi^2 = 52332.58$, $df = 22100$, $p \approx 0$; kurtosis: $b_{2,p} = 3080.08$, $z = 182.32$, $p \approx 0$). Consistent with these results, the chi-square Q–Q plot of squared Mahalanobis distances (see Appendix Figure A1) shows substantial departures from the reference line, particularly in the upper tail where observed distances are much larger than expected under χ^2_{50} . Taken together, these diagnostics reject multivariate normality for the observed item responses, which is not surprising for discrete 1–5 Likert data. This mainly cautions against treating the full raw response vector as Gaussian in downstream interpretation. Because the items are ordinal, later analyses also consider polychoric

correlations, which estimate pairwise associations under an underlying latent-response model rather than assuming that the full observed multivariate distribution is normal.

3.2 Working with ordinal data

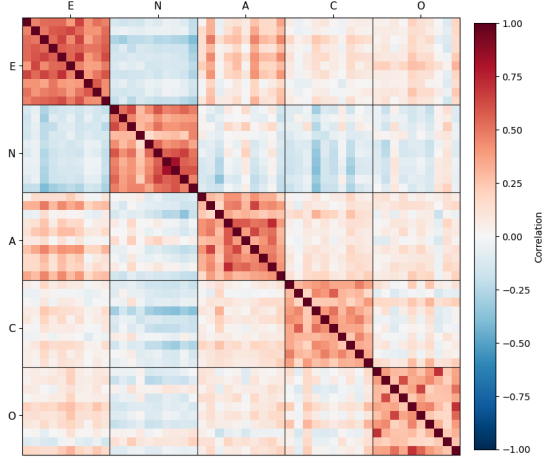
Because the Big Five items are measured on a 1–5 Likert scale, responses are ordinal and discrete rather than continuous. To better align dependence estimation with this measurement scale, I also computed polychoric correlations, which treat each item as a discretized version of an underlying continuous latent response and estimate the correlation of those latent variables.

Figure 2(a) shows the resulting 50×50 item-level polychoric correlation matrix, ordered by domain, which reveals clear within-domain blocks and comparatively weaker cross-domain associations. To provide a more interpretable summary at the domain level, Figure 2(b) aggregates these item correlations by averaging across all item pairs within and between domains (excluding self-correlations on the diagonal). The diagonal entries remain moderately large (0.34–0.51), indicating coherent within-domain structure, while most off-diagonal averages are close to zero, with the largest cross-domain averages observed for E–A (0.18) and E–N (–0.14). These polychoric-based summaries are used as the primary correlation inputs for the main factor-analytic modeling.

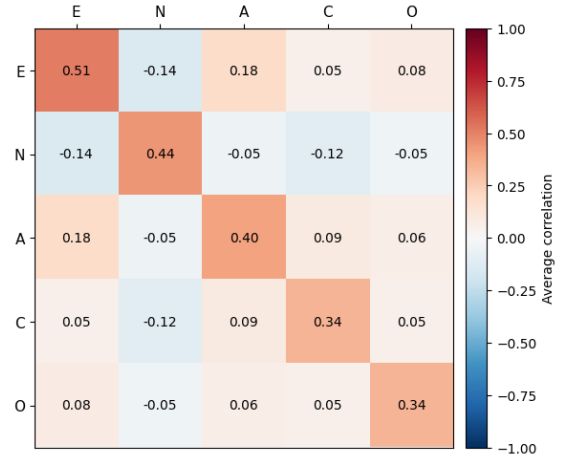
As a sensitivity check, I compared these summaries to the corresponding Pearson-correlation structure (See Appendix Figure A2(a) and (b)). The same broad block pattern appears under both choices: within-domain associations remain clearly stronger than most between-domain associations, and the largest cross-domain averages are again observed for E–A and E–N. However, the polychoric matrix is uniformly stronger within domains, with diagonal domain-level averages increasing from 0.45 to 0.51 for Extraversion, 0.40 to 0.44 for Neuroticism, 0.34 to 0.40 for Agreeableness, 0.30 to 0.34 for Conscientiousness, and 0.28 to 0.34 for Openness. The main cross-domain averages change only modestly, for example E–A increases from 0.15 to 0.18 and E–N from –0.13 to –0.14. Thus, the substantive domain-level structure is not created by the choice of correlation measure, but polychoric correlations sharpen the within-domain signal in a way that is more consistent with ordinal Likert items. For this reason, the polychoric matrix is retained for the primary factor-analysis results, while Pearson correlation PCA is used as a descriptive baseline.

3.3 Factorability: KMO

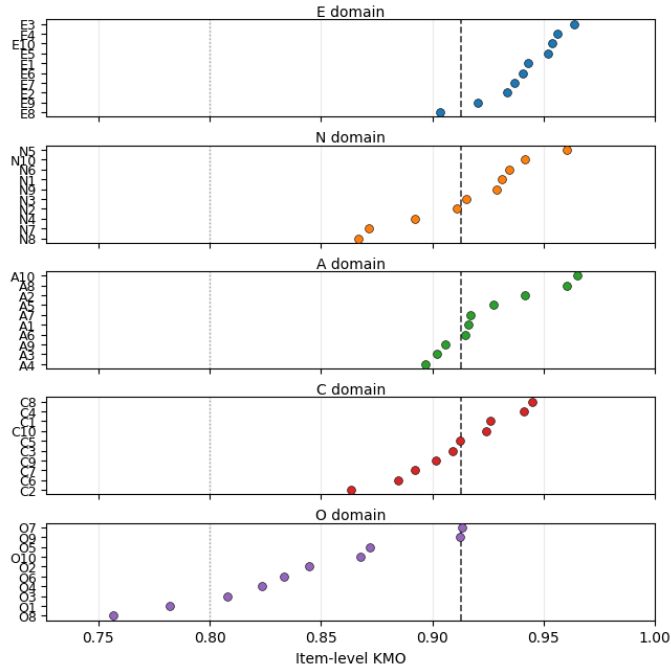
To assess whether the item set contains enough shared correlation structure to support factor extraction, I computed the Kaiser Meyer Olkin (KMO) measure of sampling adequacy. KMO compares ordinary correlations to partial correlations, with values closer to 1 indicating that a larger fraction of the variance can plausibly be attributed to common factors rather than idiosyncratic relationships. Using the polychoric correlation matrix as the input, the overall KMO was 0.91, and the item level KMO values were consistently high across domains (Figure 2(c)), with most items above 0.80 and many above 0.90. Under Kaiser’s commonly used rules of thumb, an overall KMO above 0.90 is considered “marvelous,” supporting the use of factor analysis for these data. At the same time, KMO is only a diagnostic for shared variance and can be inflated by multicollinearity, so it is interpreted here as evidence of factorability rather than a guarantee that the resulting factors will be substantively meaningful.



(a) Item-level polychoric correlation matrix (50×50), ordered by Big Five domains (E, N, A, C, O).



(b) Domain-level average polychoric correlations, computed by averaging correlations across all item pairs within and between domains (diagonal excludes self-correlations).



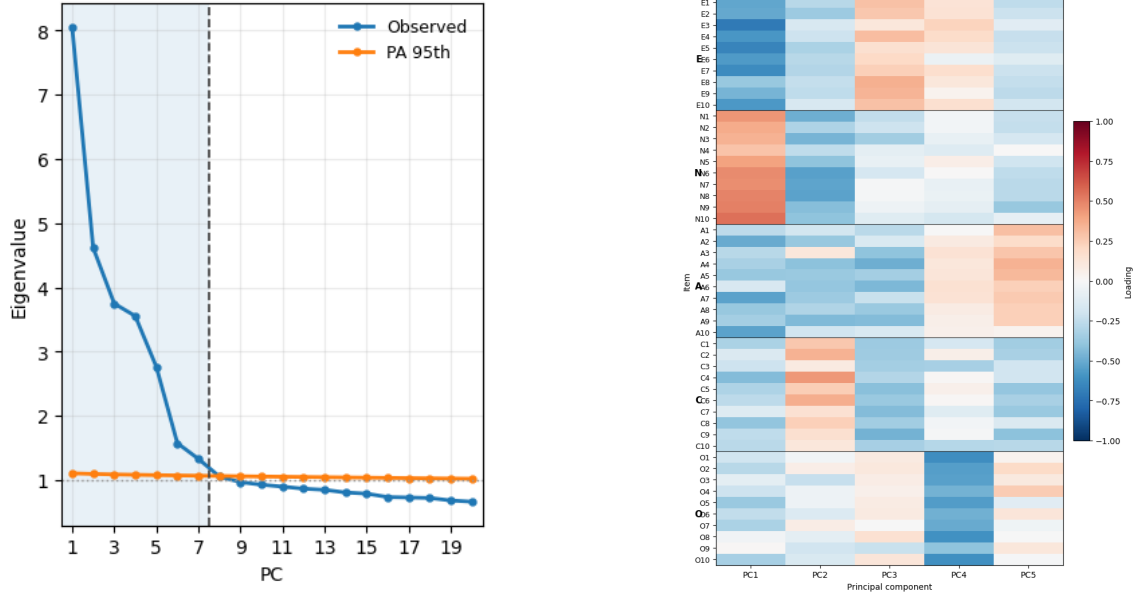
(c) Item-level KMO values by domain; the dashed line indicates the overall KMO.

Figure 2: Polychoric correlation structure and factorability diagnostics for the Big Five items after reverse-key recoding. Panels (a)–(b) summarize the dependence structure at the item level and the domain level, respectively, showing stronger within-domain coherence (0.34–0.51) and generally weak cross-domain associations (largest averages: E–A = 0.18 and E–N = −0.14). Panel (c) reports item-level KMO values, providing evidence that the correlation matrix is suitable for factor-analytic modeling.

4 PCA Baseline

As a baseline dimension-reduction method, I performed Pearson correlation PCA on the standardized, reverse-key-recoded item responses. The number of components retained was chosen using permutation parallel analysis. Figure 3(a) compares the observed eigenvalues to the 95th percentile eigenvalues from a permutation null that preserves each item’s marginal distribution while removing cross-item dependence. The observed eigenvalues exceeded the parallel-analysis threshold through PC7, so seven components were retained as the PCA baseline.

To interpret the dominant variance directions, Figure 3(b) displays loadings for PC1–PC5, with items ordered by Big Five domain. Although seven PCs were retained, interpretation focuses on the first five because PCs 6–7 lie close to the parallel-analysis cutoff and add relatively little structure. Based on the loading patterns (and supported by the domain-colored score gradients for PC1), the first five PCs can be summarized as follows: **PC1 (Extraversion-Neuroticism contrast)** shows opposite-signed, coherent loadings for E versus N items; **PC2 (Conscientiousness axis)** is dominated by C-item loadings; **PC3(Extraversion secondary axis)** captures additional within-Extraversion variation beyond PC1; **PC4(Openness-axis)** is driven primarily by O items; and **PC5 (Agreeableness axis)** shows the strongest and most consistent A-item loadings. Overall, PCA suggests a low-dimensional structure broadly consistent with the intended domains, but as a descriptive variance decomposition it does not separate common latent structure from item-specific uniqueness. I therefore use factor analysis next for the main latent-trait interpretation.



(a) Scree plot with permutation parallel analysis (95th percentile); dashed line marks the selected cutoff ($k = 7$).

(b) PCA loading heatmap for PC1–PC5 (correlation PCA), with items ordered by domain.

Figure 3: Pearson-correlation PCA baseline with permutation parallel analysis. The scree plot compares observed eigenvalues to the 95th percentile of a permutation null distribution. Although the elbow suggests a broad five-dimensional pattern, parallel analysis retains up to seven components, indicating additional secondary variance structure beyond the main Big Five domains.

5 Main method: Factor Analysis

5.1 Why factor analysis

Factor analysis (FA) serves as the primary analytical framework for this study because the IPIP Big Five items are theoretically conceptualized as multiple noisy indicators of a small set of stable latent traits. The dataset is also large relative to its dimension, with $n = 19,718$ respondents and $p = 50$ items ($n/p \approx 394$), which supports stable estimation of the correlation structure for factor analysis. This alignment with the common-factor model is supported by the high overall KMO of 0.91, which indicates that the correlation structure is suitable for factor extraction. While Mardia’s test results ($p \approx 0$) indicated significant departures from multivariate normality, a condition that might superficially favor ICA for identifying non-Gaussian sources, the decision to prioritize FA was based on a fundamental theoretical trade-off. Personality psychology posits that Big Five domains are often inherently, if weakly, correlated, as evidenced by the non-negligible polychoric associations between domains such as $E - A$ (0.18) and $E - N$ (-0.14). Consequently, the strict independence assumption required by ICA would likely mis-specify the underlying psychological structure, whereas FA allows for the recovery of a “simple structure” that more accurately reflects the overlapping variance patterns of human personality.

CCA and multivariate linear regression were also not used as primary methods because they answer a different question from the one studied here. CCA requires a meaningful partition of the variables into two blocks, but the 50 IPIP items form a single questionnaire designed to measure a common latent trait system; any two-block split would be largely artificial and would target between-block association rather than overall latent structure. Similarly, multivariate linear regression requires a natural distinction between predictors and responses, whereas these questionnaire items are symmetric indicators rather than a prediction problem with a designated outcome block. For this dataset, FA is therefore the more direct tool for recovering and interpreting shared latent traits.

5.2 Number of factors

The primary factor model was set to $m = 5$ to align with the established Big Five measurement structure and to obtain a parsimonious latent-trait representation. The rotated 5-factor loading heatmap shows a clear simple structure: each factor is dominated by a single domain block, with Extraversion items loading strongly on one factor, Neuroticism items on a second, Agreeableness items on a third, Openness items on a fourth, and Conscientiousness items on a fifth. Cross-loadings are generally small relative to the primary loadings, so each factor admits a straightforward domain-level interpretation.

Although the PCA baseline with permutation parallel analysis suggested seven variance directions above a noise baseline, PCA components can reflect secondary structure (e.g., subfacets or item-specific dependence) rather than additional broad latent traits. To assess whether extra factors improve the latent-trait representation, I fit $m = 6$ and $m = 7$ solutions as sensitivity checks using the same extraction and varimax rotation (See Appendix Figure A3(a) and (b)). In both cases, the additional factor(s) did not yield a new, well-separated domain-level block. Instead, loadings became more dispersed across domains, with weaker and less stable “primary” loadings and more cross-loadings, so the rotated solutions were harder to interpret as additional broad traits. Given the strong interpretability and simple structure of the 5-factor solution, $m = 5$ was retained as the main model, while $m = 6-7$ are treated as robustness checks rather than preferred interpretations.

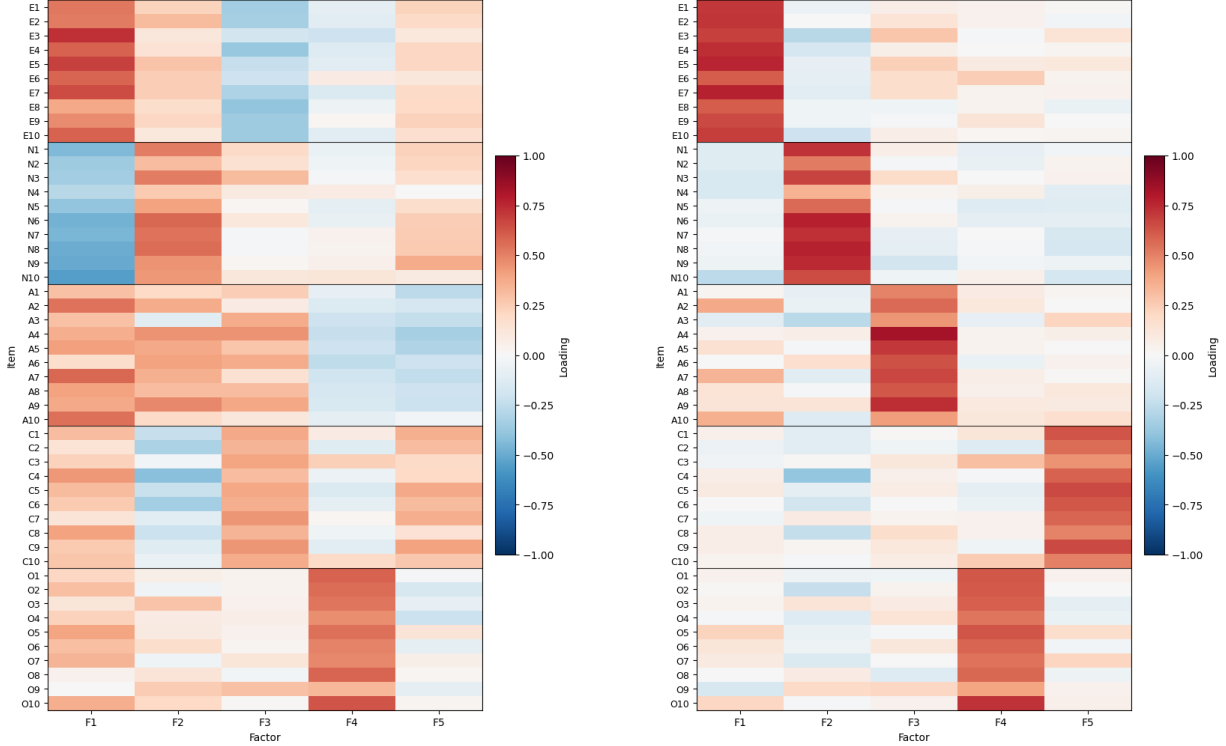
5.3 Rotation and interpretation

Because factor-analytic solutions are not unique up to rotation, the unrotated loading matrix (Figure 4(a)) should not be expected to align cleanly with the Big Five domains even when a five-factor model is appropriate. In the unrotated solution, several factors display broad, moderate loadings across multiple item blocks, and many items show non-trivial cross-loadings. As a result, the factor axes are difficult to interpret as distinct latent traits: for example, domains that should define separate traits (E, N, A, C, O) appear partially blended across factors, and the overall pattern reflects a mathematically valid basis for the common-factor subspace rather than a psychologically meaningful one.

In Figure 4(b), Varimax rotation substantially sharpens the loading structure and yields an interpretable five-factor solution that maps closely onto the Big Five domains. In the rotated heatmap, **Factor 1** is dominated by the Extraversion block: E1–E10 show consistently large positive loadings, while non-E items are near zero in comparison. **Factor 2** is dominated by Neuroticism: N1–N10 load strongly on a single factor with relatively small loadings elsewhere, forming a clean second block. **Factor 3** concentrates on Agreeableness, where A items exhibit the largest loadings and cross-domain loadings are generally weak. **Factor 4** is clearly associated with Openness: O1–O10 show strong positive loadings on this factor, with minimal spillover to other factors. Finally, **Factor 5** is dominated by Conscientiousness: the C items form the strongest loading block on this factor. Compared with the unrotated solution, varimax reduces the diffuse “moderate everywhere” pattern and instead concentrates each domain’s items onto a single factor, which is the expected simple-structure signature of a well-fitting psychometric model.

Figure 4(c) summarizes item-level communalities and uniquenesses for the Extraversion block, which quantify how much of each item’s variance is explained by the common factors versus item-specific variance. For example, item E5 (“Take charge”) exhibited the highest communality at 0.665, indicating that approximately 66.5% of its variance is explained by the latent factor structure. This identifies E5 as a core indicator of the Extraversion domain with high factorial validity and relatively low measurement noise. Conversely, item E1 (“Don’t talk a lot”) showed a higher uniqueness of 0.494, suggesting that nearly half of its variance remains uncaptured by the five broad traits. This discrepancy implies that while the overall factor structure is stable, specific items may contain linguistic nuances or relate to narrower sub-facets of sociability that the broad Big Five model does not fully parameterize. Nevertheless, since the displayed Extraversion items maintained communalities above 0.50, the results suggest that the majority of observed variance is indeed driven by the shared latent trait system rather than idiosyncratic error.

As a robustness check, I also fit the same 5-factor model using promax rotation (See Appendix Figure A4), with the same domain level alignment and no material changes in item grouping. Together with the $m = 6$ and $m = 7$ sensitivity checks above, this suggests that the substantive five factor interpretation is reasonably robust to nearby methodological choices. This indicates that the substantive interpretation of the factor structure is not sensitive to the choice between orthogonal (varimax) and oblique (promax) rotation for this dataset; promax mainly relaxes the orthogonality constraint while preserving the same underlying trait interpretation.



(a) Unrotated MINRES factor loadings for the $m = 5$ model fit to the polychoric correlation matrix. Items are ordered by domain (E, N, A, C, O); colors indicate loading sign and magnitude.

(b) Varimax-rotated MINRES factor loadings for the same $m = 5$ model. Rotation yields a clearer simple structure, concentrating large loadings within domains.

(c) Varimax-rotated factor analysis results for the Extraversion items under the $m = 5$ specification, showing item-level communality and uniqueness.

Item	Communality	Uniqueness
E1	0.506	0.494
E2	0.530	0.470
E3	0.638	0.362
E4	0.572	0.428
E5	0.665	0.335

Figure 4: Factor-analysis results for the Big Five items under a 5-factor MINRES model fit to the polychoric correlation matrix. Panel (a) shows the unrotated loading matrix, where factors are harder to interpret due to mixed domain loadings. Panel (b) shows the varimax-rotated loadings used for interpretation, yielding a clearer simple structure aligned with the five domains (E, N, A, C, O). Panel (c) reports representative communalities and uniquenesses, illustrating how much item variance is captured by the common factors versus item-specific variation.

6 Discussion and Conclusion

Overall, the results support a stable five factor latent structure that is broadly consistent with the intended Big Five domains when the ordinal nature of the data is taken into account through poly-

choric correlations and rotated factor solutions. Although PCA and parallel analysis suggested up to seven variance directions above a noise baseline, only five factors remained clearly interpretable as broad personality domains under factor analysis. The additional PCA directions are therefore better interpreted as secondary variance structure rather than as two extra substantive traits.

The comparison between PCA and FA is informative rather than contradictory. PCA summarizes dominant variance directions, whereas FA is more appropriate for the present goal because it separates shared covariance from item specific variance through uniqueness. Rotation was also important for interpretation: the unrotated FA solution mixed domains, while varimax yielded a much clearer simple structure, and promax produced a similar domain mapping. This suggests that the main five domain interpretation is robust to allowing correlated factors.

The main conclusion was also stable across Pearson versus polychoric correlations and across nearby choices of the factor number. In particular, solutions with $m = 5$, $m = 6$, and $m = 7$ indicated that additional factors mainly captured narrower residual structure rather than clearly separated new personality domains. FA was preferred to ICA because these items are intended to measure correlated latent traits rather than statistically independent sources, while CCA and multivariate linear regression were not appropriate because the 50 items do not form a natural two block or predictor response structure.

Several limitations remain. The responses are ordinal and strongly non normal, so polychoric correlations still rely on an underlying latent continuous approximation. In addition, the sample is drawn from an online self selected setting, and minor wording related residual dependence may remain even after reverse coding. Finally, uncertainty was assessed mainly through sensitivity analyses rather than formal resampling based intervals.

Taken together, the evidence favors a five factor representation of the questionnaire, with modest residual item level structure beyond the idealized simple domain pattern.

References

- [1] Goldberg, Lewis R. (1993). The structure of phenotypic personality traits. *American Psychologist*, 48(1), 26–34.
- [2] Goldberg, Lewis R. (1999). A broad-bandwidth, public-domain, personality inventory measuring the lower-level facets of several five-factor models. In *Personality Psychology in Europe*, Vol. 7, pp. 7–28. Tilburg University Press.

Appendix

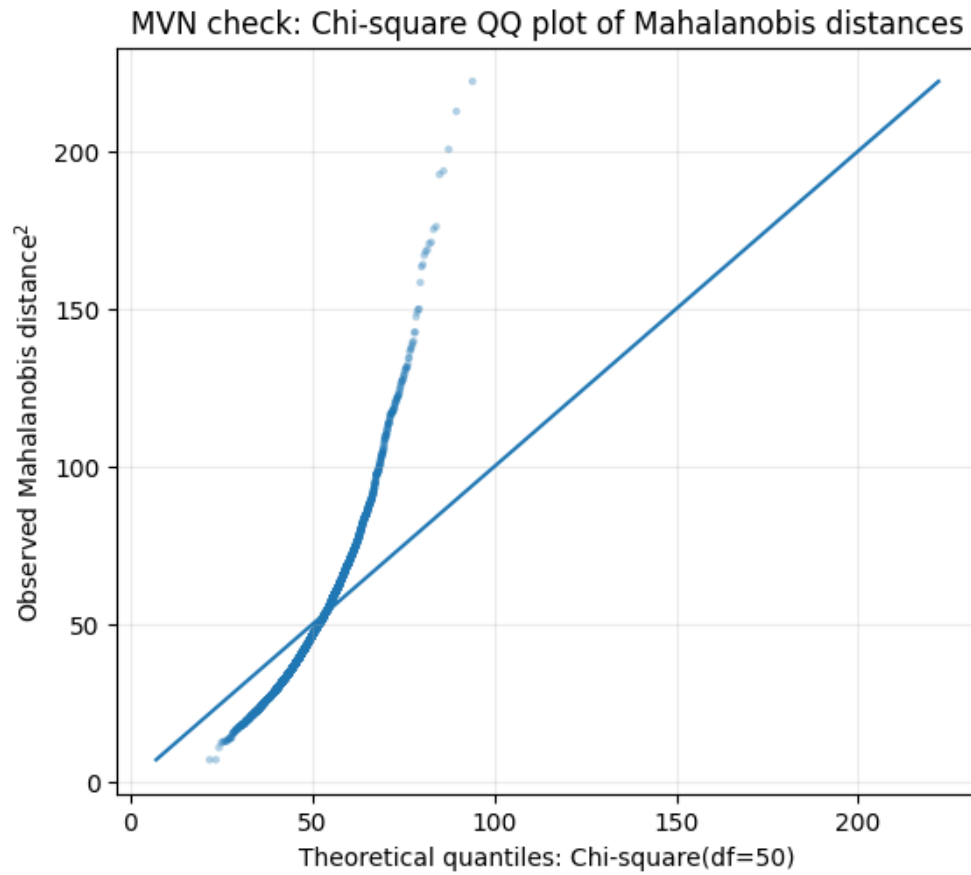
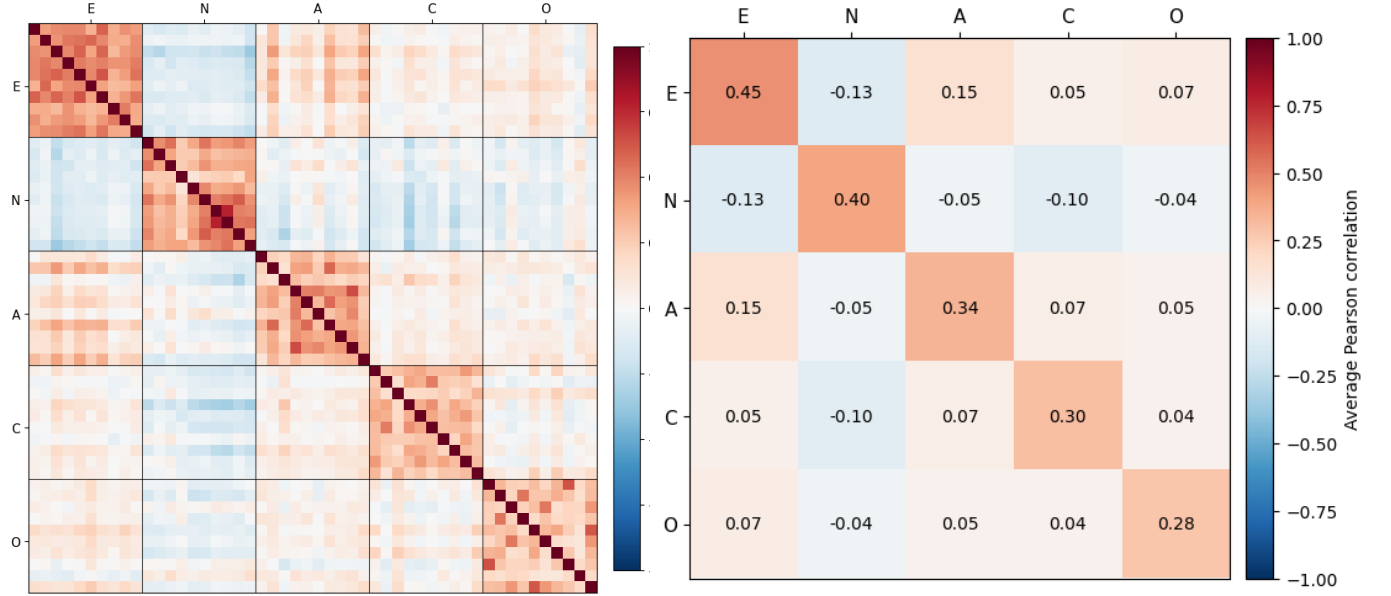


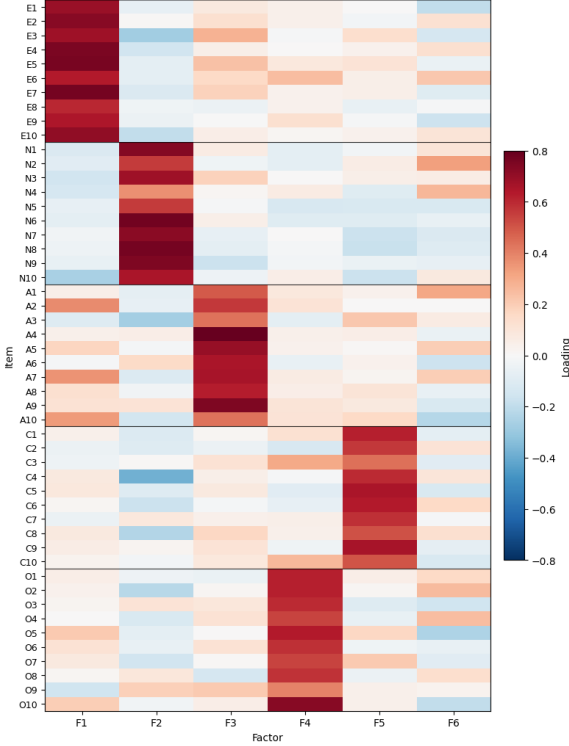
Figure A1: Chi-square Q–Q plot of squared Mahalanobis distances for the 50 standardized Big Five items. Under multivariate normality, the points would fall approximately along the reference line. Instead, the pronounced upward deviation in the upper tail indicates substantially larger observed distances than expected under χ^2_{50} , providing visual evidence against the MVN assumption.



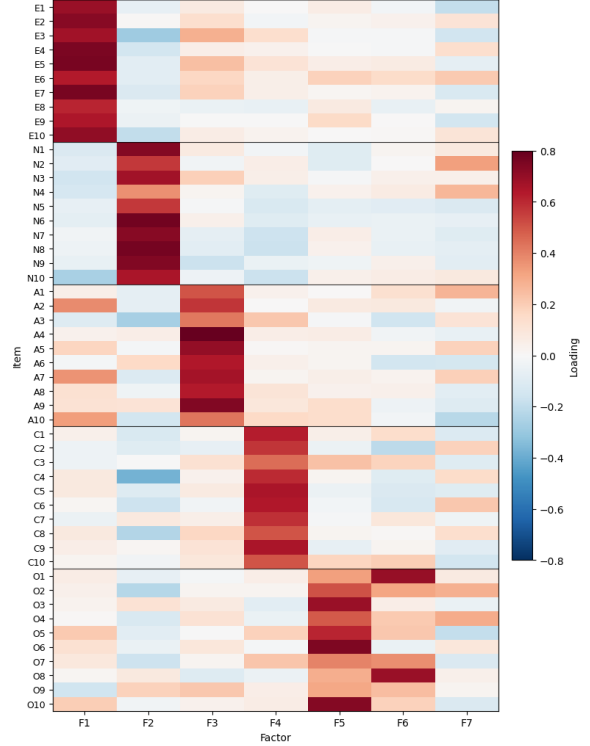
(a) Item level Pearson correlation matrix for the 50 IPIP Big Five items, ordered by domain after reverse key recoding. The strongest positive correlations appear within the same domain blocks, while most between domain correlations are weaker, indicating that the broad Big Five structure is already visible under Pearson correlations.

(b) Domain averaged Pearson correlations obtained by averaging item correlations within and between the five Big Five domains. Within domain averages are clearly larger than most cross domain averages, with the strongest positive cross domain average between Extraversion and Agreeableness and the strongest negative average between Extraversion and Neuroticism.

Figure A2: Pearson correlation sensitivity check after reverse key recoding. Panel (a) shows the item level Pearson correlation matrix and panel (b) shows the corresponding domain averaged correlations. The same broad domain structure appears as in the main polychoric analysis, although within domain associations are somewhat weaker, indicating that the qualitative conclusions are robust while polychoric correlations remain preferable for the primary analysis of ordinal Likert items.



(a) Varimax-rotated loadings, $m=6$. Loadings from a 6-factor MINRES solution fit to the polychoric correlation matrix (items ordered by domain). The additional factor does not form a distinct new domain-level block; instead, loadings are more dispersed with increased cross-loadings.



(b) Varimax-rotated loadings, $m=7$. Loadings from a 7-factor MINRES solution fit to the polychoric correlation matrix (items ordered by domain). Splitting into more factors further fragments the structure, with weaker separation of domain blocks and less clear simple structure.

Figure A3: Sensitivity of the factor solution to the number of factors beyond the Big Five benchmark. Both the 6- and 7-factor varimax solutions are shown using the same loading scale and item ordering to facilitate comparison. Relative to the 5-factor model, increasing m does not yield an additional clean, interpretable trait-like factor; instead, the extra factors primarily redistribute smaller loadings across domains, reducing interpretability and suggesting they capture residual structure rather than an additional broad latent trait.

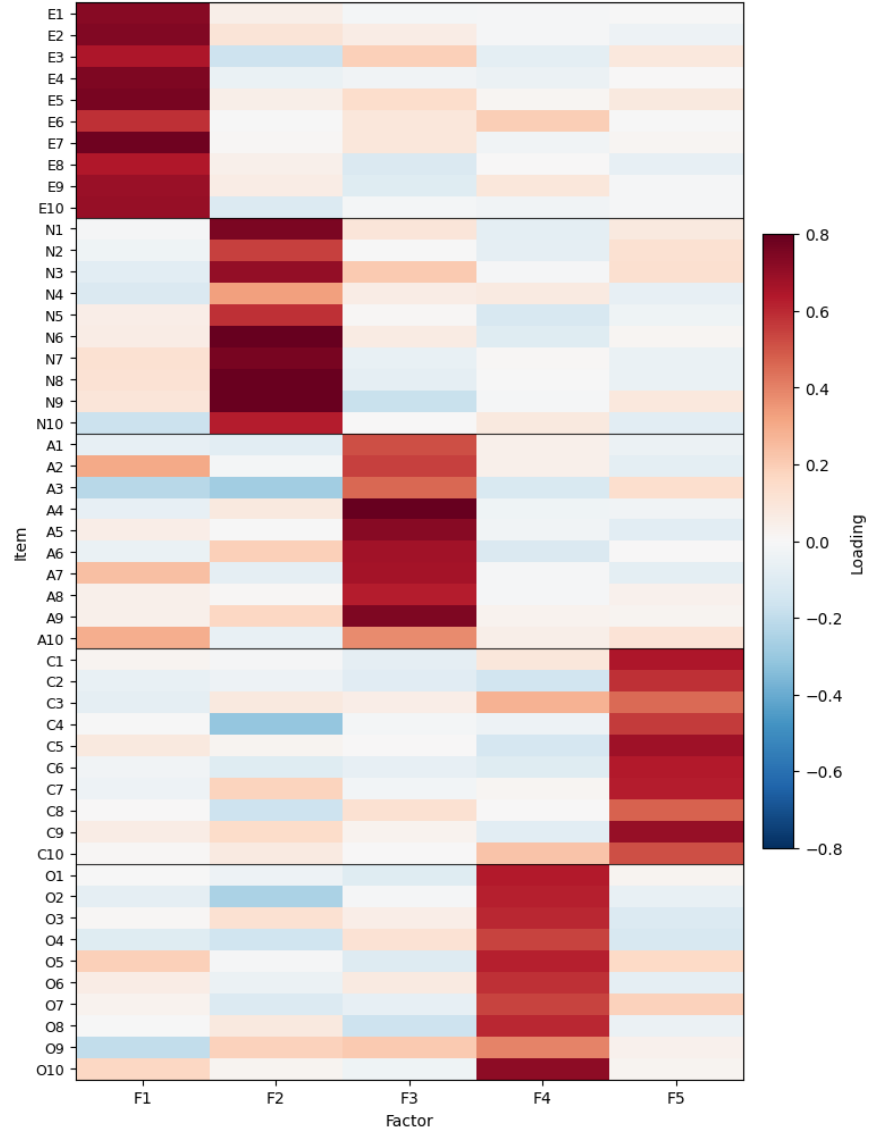


Figure A4: Promax-rotated factor loadings for the $m = 5$ MINRES solution fit to the polychoric correlation matrix. Items are ordered by Big Five domain (E, N, A, C, O), and the color scale shows loading sign and magnitude. Compared with the orthogonal varimax solution, promax allows correlated factors but yields a qualitatively similar simple structure, supporting the robustness of the five-factor domain interpretation to the choice of rotation.